

Task 1 — Build & Evaluate a Linear Regression Model

Project: House Price Predictor | Submitted by: Bishwajeet Kumar | MainCrafts Technology

1. Objective

The goal of Task 1 was to introduce the complete Machine Learning workflow by building and evaluating a Linear Regression model on the California Housing Dataset. This project covers data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation using standard metrics, and result visualization — forming a strong foundation for all subsequent ML tasks.

2. Dataset Overview

2.1 California Housing Dataset

The California Housing Dataset, built into scikit-learn, contains data from the 1990 California census. It is a widely used benchmark dataset for regression problems.

Property	Details
Total Samples	20,640
Input Features	8
Target Variable	MedHouseVal (Median House Value in \$100,000s)
Missing Values	None
Source	sklearn.datasets.fetch_california_housing

2.2 Input Features

- **MedInc** — Median income in block group — strongest predictor of house price
- **HouseAge** — Median house age in block group
- **AveRooms** — Average number of rooms per household
- **AveBedrms** — Average number of bedrooms per household
- **Population** — Block group population
- **AveOccup** — Average number of household members
- **Latitude** — Block group latitude (geographic location)
- **Longitude** — Block group longitude (geographic location)

3. Exploratory Data Analysis (EDA)

3.1 Key Findings

- No missing values found in the dataset — no imputation required.
- MedHouseVal (target) is right-skewed with values capped at 5.0 (\$500,000).
- MedInc has the highest positive correlation with house prices ($r \approx 0.69$).
- Latitude and Longitude show negative correlations — location plays a major role.
- AveRooms and AveBedrms are moderately correlated with each other (multicollinearity).
- Population and AveOccup have very low correlation with house prices.

3.2 Correlation Summary

Feature	Correlation with MedHouseVal	Direction
MedInc	0.688	Strong Positive
AveRooms	0.151	Weak Positive
HouseAge	0.106	Weak Positive
AveBedrms	0.046	Very Weak Positive
Population	-0.025	Very Weak Negative
AveOccup	-0.023	Very Weak Negative
Longitude	-0.045	Weak Negative
Latitude	-0.144	Weak Negative

4. Model Training

4.1 Preprocessing & Split

- All 8 features were used as input (no feature selection required at this stage).
- Data split: 80% training (16,512 samples) and 20% test (4,128 samples).
- `random_state=42` used for reproducibility.
- No feature scaling applied in Task 1 (addressed in Task 2).

4.2 Linear Regression Model

A standard Ordinary Least Squares (OLS) Linear Regression model was trained using scikit-learn's `LinearRegression` class. The model learns a linear relationship between the input features and the target variable by minimizing the sum of squared residuals.

5. Model Evaluation Results

5.1 Performance Metrics

Metric	Value	Meaning
MAE (Mean Absolute Error)	~0.5332	Avg error of \$53,320 per prediction
RMSE (Root Mean Squared Error)	~0.7255	Penalizes large errors more
R ² Score	~0.576	Model explains 57.6% of variance

5.2 Visual Analysis

- Actual vs Predicted plot: Points cluster near the reference line for low-to-mid price ranges, but scatter increases for higher values — indicating the model struggles with expensive properties.
- Residuals plot: Residuals are roughly centered around zero, confirming no major systematic bias. Some heteroscedasticity is visible at higher predicted values.
- Feature coefficients: MedInc has the largest positive coefficient, confirming its dominance as a predictor. Latitude and Longitude have negative coefficients reflecting geographic pricing trends.

6. Improvement Ideas

- Apply StandardScaler for feature normalization — improves model stability (implemented in Task 2).
- Try Ridge or Lasso Regression to reduce overfitting via regularization.
- Use ensemble models like Random Forest or Gradient Boosting for better accuracy.
- Engineer new features such as rooms-per-person or income-to-age ratio.
- Remove or cap outliers in Population and AveOccup to reduce noise.
- Apply log transformation to the skewed target variable (MedHouseVal).

7. Conclusion

Task 1 successfully introduced the complete ML workflow — from raw data loading and EDA to model training, evaluation, and visualization. The Linear Regression model achieved an R² of ~0.576, explaining over half the variance in California house prices. MedInc (Median Income) was identified as the strongest predictor. While the baseline results are reasonable, the improvement ideas above provide a clear roadmap for building better models — explored in Tasks 2 and 3.